

SHARED PREPROCESSING

-  **arXiv LaTeX sources**
publicly available source packages
-  **parse and clean**
truncate appendix, drop figures
-  **structural filter**
body length, section count
-  **deduplicate**
content hash, drop benchmark leakage

(a) CPT data

papers 2006 – 2026.01



quality scoring

small LLM, low threshold

1.8M papers



unfold to trajectories

Qwen3.5-4B / 9B / 27B

60B tokens

one corpus per model

(b) SFT data

papers 2006 – 2026.01



stricter filtering

strong LLM, high bar

356K papers



balance 8 fields

cap the largest fields

218K papers



reframe as tasks

Qwen3.6-27B

650K drafts



rule check and judge

drop unsupported claims



200K samples

29 task types

(c) PAW-Bench

papers 2026.02 – 2026.06



balanced sampling

month x subject category

2,000 papers



quality and fit

top 50% within each field

735 papers



pose questions

strong LLM



audit and repair

manual spot check



2,940 tasks

15 task types